



Packet Capture Analysis:

I have analyzed the provided packet capture file using the free network analysis tool Wireshark. I was able to put "http" into the filter field in order to filter the network traffic to only see HTTP packets.

This view let me see some interesting http GET requests, which indicate that the user specifically requests information.

Sub-task 1:

To find the images the user accessed called anz-logo.jpg and bank-card.jpg I followed the following process for both images:

First I filtered the packet capture for http traffic and looked through the remaining packets for the GET request that downloaded the image. I then right clicked the image and followed its TCP stream. In the TCP stream I saw what looked like image data. In order to view the data in hex format, I changed the view to „raw“, and then searched the hex data for a jpeg's file signature.

After finding the file signature "FFD8" at the top, and the file footer "FFD9" at the bottom, I copied everything between those two points into the hex editor HxD and saved it as a jpg image. Resulting in the image below.



Bank-logo.jpg



Bank-card.jpg

Sub-task 2:

I followed the same process to extract these images as I did in sub-task 1, which was to view the TCP stream, identify the images hex data, then copy and save that as a jpg file.

The image for ANZ1.jpg :

PROTECT YOUR VIRTUAL VALUABLES

TAKE SOME SIMPLE STEPS TO
PROTECT YOUR INFORMATION



 ANZ Cyber Secure



ANZ1.jpg

The difference in the network traffic for this images download I discovered was a hidden message in the data after the end of the image.

The message said "You've found a hidden message in this file! Include it in your write up."

MAKE A 'PACT'

TO PROTECT YOUR VIRTUAL VALUABLES



PAUSE
before sharing your
personal information

Ask yourself, do I really need to give my information to this website or this person? If it doesn't feel right, don't share it.



CALL OUT
suspicious messages

Be aware of current scams. If an email, call or SMS seems unusual, check it through official contact points or report it.



ACTIVATE
two layers of security with
two-factor authentication

Use two-factor authentication for an extra layer of security to keep your personal information safe.



TURN ON
automatic
software updates

Set your software, operating system and apps to auto update to make sure you get the latest security features.

Report suspicious messages from ANZ:

✉ Email hoax@cybersecurity.anz.com

Report fraudulent or unusual ANZ account activity:

☎ 137 028 / +61 3 8693 7153 (Corporate/Business Clients)

☎ 133 350 / +61 3 9683 8833 (Personal Banking Customers)

Australia and New Zealand Banking Group Limited (ANZ) ABN 11 005 357 522. Item No. 96184B 09.2018 AU22349

ANZ2.jpg

This network traffic also had a message hidden in the same way.

It was "You've found the hidden message! Images are sometimes more than they appear."

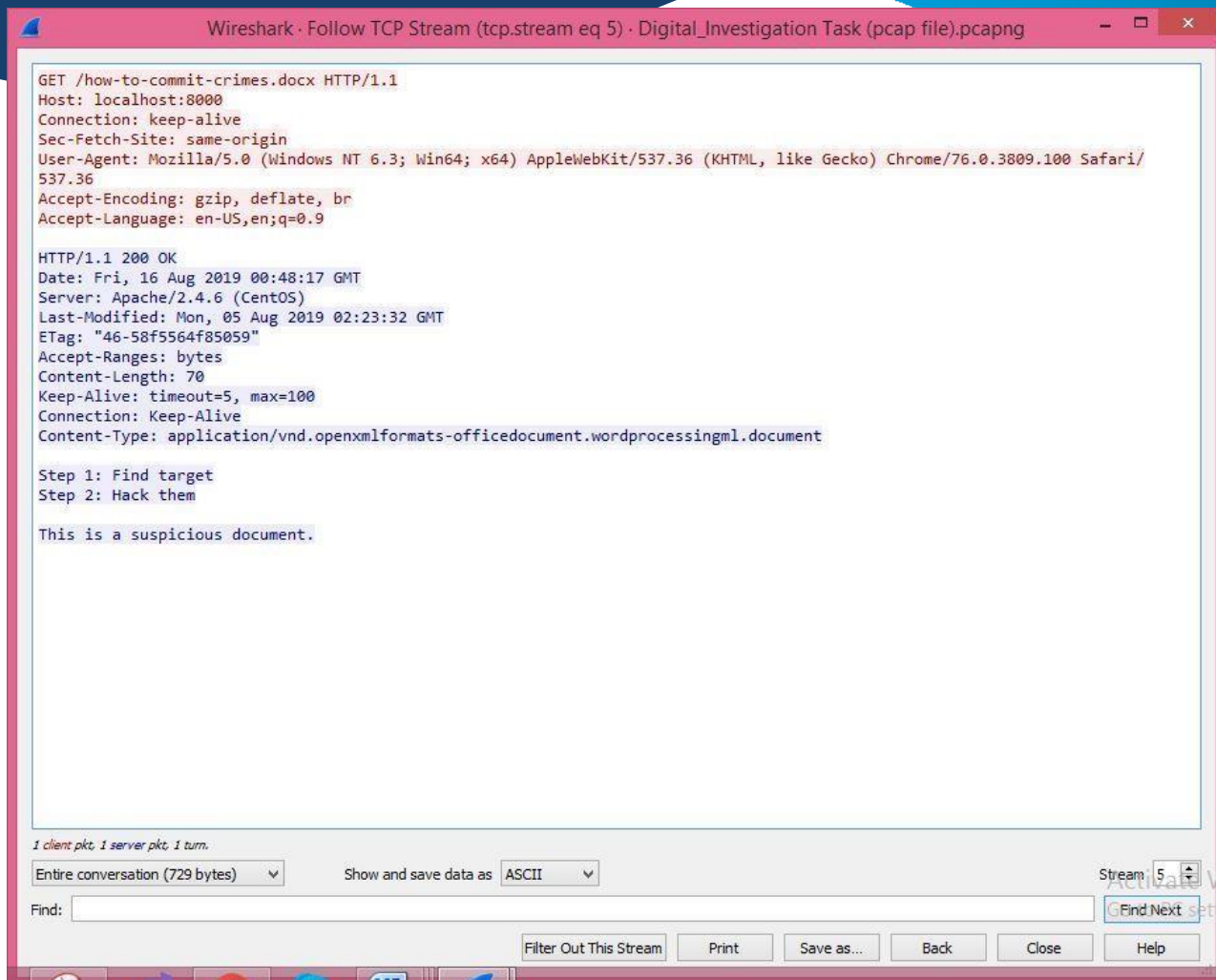
Sub-task 3:

In order to find the contents of the document, I had to view the TCP stream of the http get request for the file. The documents contents were visible in the ASCII view.

Step 1: Find target

Step 2: Hack them

This is a suspicious document



Sub-task 4:

In order to view these PDF's I viewed the TCP stream as usual, and found the file signature for a PDF, which was the hex data "25 50 44 46". I noticed in the ASCII view that the PDF data went until the very end of the TCP stream, so I copied all the hex data from the file signature onwards into HxD and saved it as a pdf file.

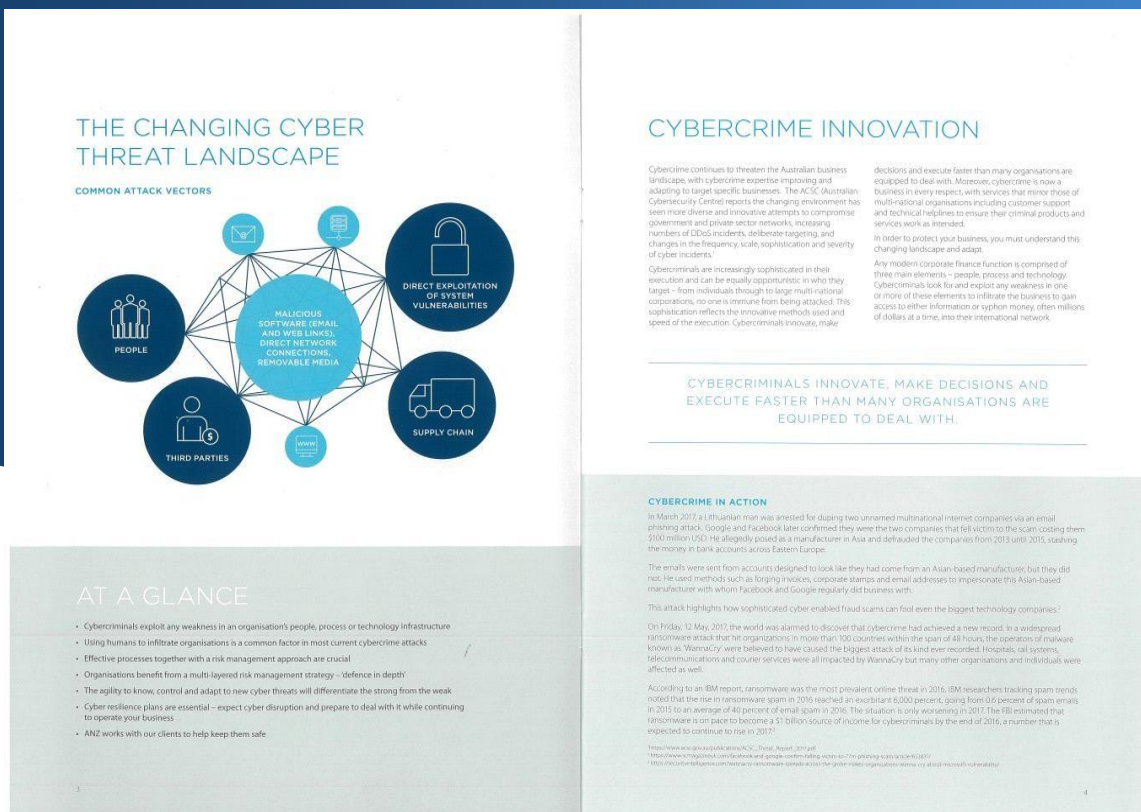
The same process worked for all three files:

VOLUME
2

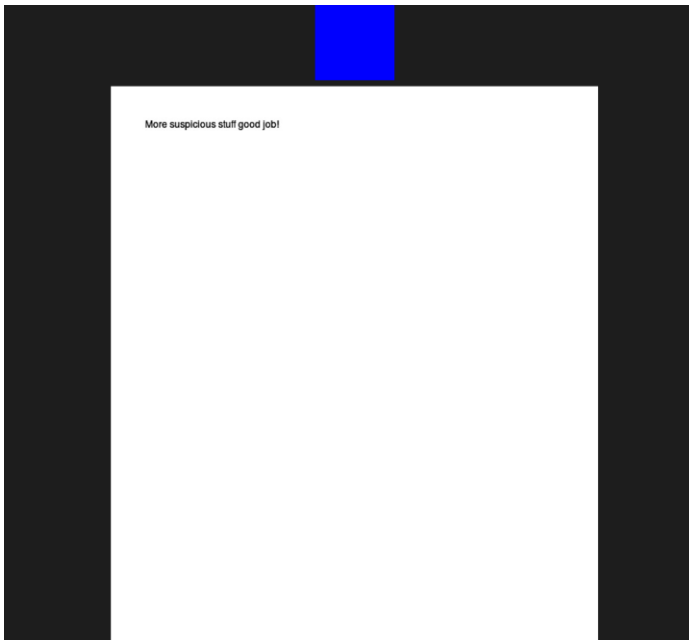
CYBERSECURITY

WORKING TOGETHER TO KEEP YOUR
ORGANISATION SAFE





ANZ_document2.pdf (extracted image)



Evil.pdf(extracted image)

Sub-task 5:

I viewed the TCP stream of this file, and noticed that instead of being plain text it was encoded data and when viewed as hex it had the same file signature as a jpg image. So I copied and saved the hex data with HxD as I have for other images, and discovered that the text file was actually this image.



Sub-task 6:

I viewed the TCP stream as normal when investigating this traffic, and found two sets of jpeg file signatures. In the TCP stream I saw what looked like image data. In order to view the data in hex format, I changed the view to „raw“, and then searched the hex data for a jpeg’s file signature. After finding the file signature “FFD8” the top, and the file footer “FFD9” at the bottom, I copied everything between those two points into the hex editor HxD and saved it as a jpg image.

I tried extracting both sets of data, and got two different images. Resulting in the image below.

First image:



Second image:



So the thing that is different about this traffic is that a single GET request performed by the user downloaded two images.

Sub-task 7:

To find the images the user accessed called *broken.png* I followed the following process for both images:

First I filtered the packet capture for http traffic and looked through the remaining packets for the GET request that downloaded the image. I then right clicked the image and followed its TCP stream. In the TCP stream I saw what looked like image data. In order to view the data in hex format, I changed the view to „raw“, and then searched the hex data for a jpeg's file signature. After finding the file signature "89 50 4e 47 0d 0a 1a 0a" I copied everything after that point to end and then copy into the hex editor HxD and saved it as a png image. The image as follow:



Sub-task 8:

After investigating TCP stream for securepdf.pdf I discover following thing:

The data there was not for a PDF. The bottom of the file contained the hidden message: Password is "secure"
It contained the file signature for a zip file, meaning that the the user downloaded was actually a zip file.

So I copied the hex of the zip file into HxD and saved it as a zip file. I opened this zip file, and found it contained a pdf file called rawpdf.pdf. When opened, the pdf prompted for a password. The password „secure“ shown in the tcp stream worked and the PDF opened. It was the first two pages to a guide for internet banking.



YOUR GUIDE TO ANZ INTERNET BANKING



TABLE OF CONTENTS

Why use ANZ Internet Banking?	3
Online Security	4
Getting started	5
Viewing your accounts	6
Transferring funds	7
Check the details before you pay	8
Your transfer receipt	9
Paying bills	10
Using Pay Anyone	11
International Money Transfers	12
Logging Off	13
Things you need to know	14
Frequently asked questions	15